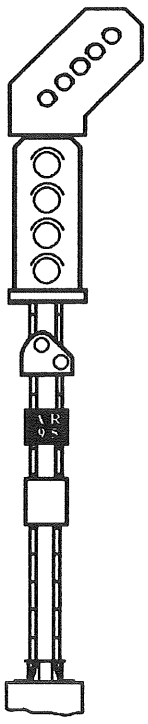
INTRODUCTION TO SIGNALLING

6

LINESIDE SIGNALS



A.J.R.Rowbotham

INTRODUCTION

Signals, positioned along the trackside, pass information to the drivers of passing trains.

The signals might well be mechanical, where the different physical positions of an arm give different indications, or light signals where different lights give different indications. Mechanical signals will not be discussed in this document.

Different indications, or *aspects*, may be given in different ways. Either different colours, (or combinations of colours), or a different pattern of lights (with the actual colours not necessarily relevant) may be used. The former is known generally as a *colour light signal* and the latter is known as a *position light signal.*

Throughout the world three colours are universally used, although they may be used in different ways in different countries:

- Red Stop.
- Green Proceed.
- Yellow Caution (Yellow is never called 'Amber' in railway signalling).

Other colours are found, particularly White and in some countries, Blue. Blue light has very poor propagation properties through the atmosphere, so can only be used where it does not have to be seen from a distance.

On most railways train movements are divided into two types:

Main movements	Trains carrying out their principle task of moving passengers or freight from A to B.
Shunt movements	Ancillary movements such as moving engines onto the front of a train, or into the engine shed, moving empty carriages into a station, backing a freight train into a siding etc.

The stop aspect of a signal is said to be the *most restrictive* aspect, and the proceed aspect is said to be the *least restrictive* aspect. A caution aspect is between the two, and is consequently *less restrictive* than a stop aspect and *more restrictive* than a proceed aspect.

When a signal changes from a stop aspect to a less restrictive aspect it is said to *clear.* Other important terms used in connection with signals are *on* and *off.* A signal showing its most restrictive aspect is said to be 'on', and when it clears (to any other aspect) is said to be 'off'. These abstract terms are left over from the last century. In France more meaningful words are used for On and Off, 'closed' and 'open' respectively.

In the UK, colour light signals are used for main moves, and position light signals are used for shunt moves. Colour light signals are controlled by the interlocking so they give full safety protection when cleared, whilst shunt signals do not necessarily give equally full protection.

There are, throughout the world, two basic signalling strategies, *Speed Signalling*, and *Route Signalling*.

With speed signalling systems, the signals give the driver additional information about the range of speeds at which he should drive. In some systems speeds are defined merely as 'low', 'medium' or 'high' and the aspects are defined in these terms. This leads to a system

needing perhaps 14 or 15 different aspects to be differentiated by the driver. In more modern systems, which tend to be simpler, the actual speed is indicated to the driver numerically. In the UK there is a strict rule that drivers should never be required to pass a red aspect at speed. In some speed signalling systems this would be an every day experience - one aspect might well be a signal showing red, yellow and green at the same time.

With route signalling it is up to the driver to drive at the correct speed, using his intimate knowledge of the line and its speed limits. (Drivers are only allowed to drive over routes for which they hold a certificate.). At junctions it is only necessary to indicate to the driver that a diverging route is set. As a result, route signalling leads to a much simpler system of aspects. Route signalling is used in the UK.

COLOUR LIGHT SIGNALS

The Colour light *signalhead* consists of separate light proof compartments each holding a lamp, lampholder and lens. The lenses are suitably coloured to produce red, yellow or green light, taking into account the yellowish light emitted by the lamps. The lamp must be accurately placed with respect to the lens to ensure efficient performance.

Mirrors must not be used to enhance the light output. This is to prevent the danger of sun light entering the signal head, reflecting back from the mirror and appearing to the driver to be a green aspect. The inside of the signal head is painted matt black to similarly eliminate any chance of reflections.

Each aspect has a *hood* to stop the sun shining directly on the lens, and each signal has a *background* to make it easier to spot from a distance.

Symbols

Signals need to be shown on track plans, so agreed symbols (defined in BS 376) are used. Each lamp is represented by a circle, and bars in the circle indicate the colour. A horizontal line, barring the way, is used for a red aspect, a vertical line for the green and a diagonal line for yellow.

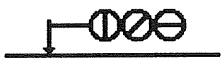


These symbols may be combined together to represent a signal, for instance a three aspect signal, made up of a red, a yellow and a green aspect would have the three lamp symbols arranged vertically on top of a 'post'.



The most restrictive aspect of a physical signal has to be placed nearest the drivers' eye level, and this normally means the red is the lowest aspect. The yellow is normally placed between the red and the green.

On track plans the railway lines normally run horizontally across the page rather than vertically, so the symbol just drawn has to be rotated through 90°, resulting in the post of the signal being shown parallel to the line. It is also necessary to draw a short arrow from the base of the post to the line to which the signal applies, otherwise in congested areas of track confusion will ensue. Signals are usually physically placed on the left hand side of the line to which they apply. It may be that because of physical obstruction a signal so placed may not be visible from an adequate distance (the *sighting* is not adequate), in which case it can be placed on the right hand side of the track if the sighting is then improved. A conventionally placed three aspect signal would be shown



This is a signal for traffic moving from left to right, i.e. the bottom of the post is met first in the direction of travel. Note also that the signal has to be viewed from the track to determine the colours represented by the symbols. The lowest lamp is still the red aspect, the bar representing the colour having also been turned through 90°.

A signal for traffic moving from right to left, conventionally placed on the left hand side of the line is shown



If the above signals were placed on the right hand side of the line, the exception rather than the rule, then the symbols would be, respectively



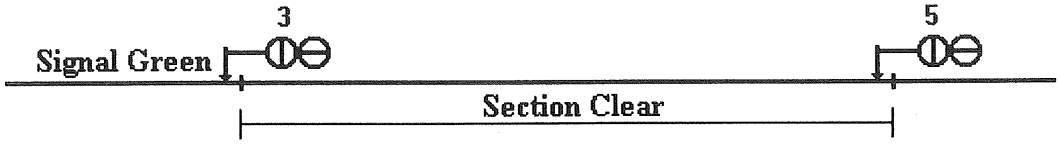


2-aspect signalling

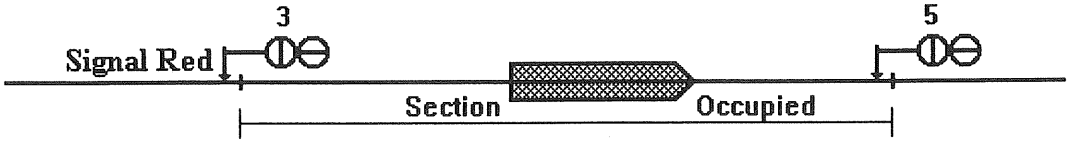
The simplest signalling system would consist of 2-aspect signals, that is signals showing only a red or a green aspect. Consider a section of line with 2-aspect signals, numbers 3 and 5. (Unless otherwise stated traffic movement in the examples will always be from left to right)



If the section of track between signals 3 and 5 is not occupied by any train, then it would seem reasonable that signal 3 could show a green aspect.



However, if there is a train between the two signals, then signal 3 must show a red aspect.



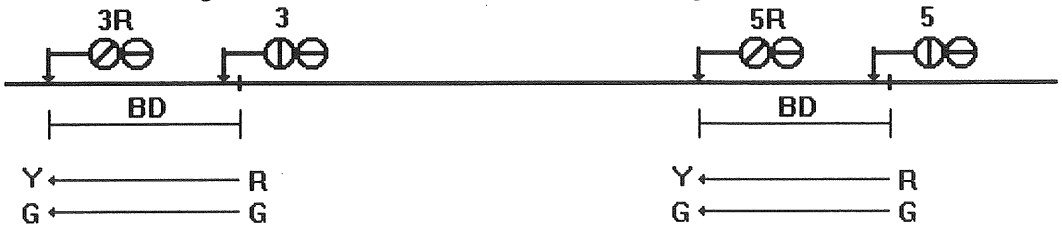
At first sight this might seem to be a satisfactory basic signalling system - 2-aspect signals placed along the track, with each allowed to show a green aspect if the section of track up to the next signal is clear. This system ensures that only one train at a time can be between each pair of adjacent signals.

However, on further consideration it is not very practical. If a train was to approach signal 3 at 150 km/h when it was showing a red aspect, by the time the driver had seen the signal there would be nowhere near enough time or space to actually stop at the signal. (It would take perhaps 2½ km to stop such a train.) Consequently, some sort of warning must be given to the driver that he is approaching a red aspect, and this is done by providing *repeater signals.*



Each red/green signal (*Stop signal*) now has a yellow/green repeater (with the suffix R in its number). Each repeater is arranged to show yellow (Y) if its parent stop signal is showing red (R), and green (G) if its stop signal is also showing green.

Each repeater must be placed at such a distance from its stop signal that the heaviest fastest train on the line has adequate distance to come to a halt at the stop signal, having sighted the yellow repeater signal aspect. This distance is known as the *Braking Distance (BD)* and has to take into account the gradient of the line between the two signals.

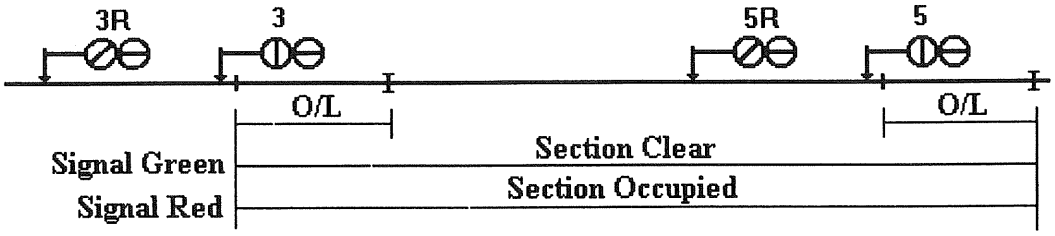


Consider another situation with a stationary train <u>just</u> past a stop signal



Under this condition 5 will be red, 5R yellow, 3 and 3R green. The driver of a following train will see the yellow aspect on 5R and start braking. If he misjudges the braking (for instance, the train might be slightly heavier than usual or there might be the wrong sort of leaves on the line) the train might just edge past the signal and collide with the stationary train.

This problem is solved by only allowing a stop signal to clear if the section between it and the next signal plus a further short distance, known as the *overlap (O/L)*, is clear. This overlap just gives the driver a little extra leeway to allow for minor misjudgement.



Note that a train situated just past a signal as in the previous diagram but one, will ensure that both 5 and 3 signals are red, 3 being red because the train is occupying the overlap area at 5 signal. Thus, as a train moves along the line, it is protected by two red signals behind it, whilst it is occupying the overlap of each signal.

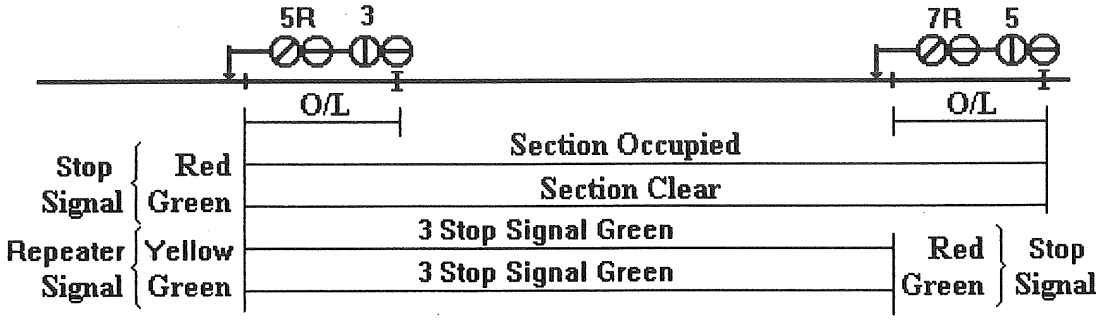
The distance between the stop and repeater signal must not be less than the BD, but the distance between successive stop signals is not fixed but determines the *Headway* of the line. The headway is a measure of the capacity of the line and can be measured either as the number of trains per hour, or the time interval between successive trains. As was seen above, only one train can be between a pair of stop signals at a time. If it takes six minutes for a train to cover this distance then only ten trains an hour are possible (there are ten lots of 6 minutes in an hour). To improve the headway the stop signals need to be closer together, but the stop/repeater distance must not alter, so they must move as a pair.

This basic 2-aspect system might well be found on lightly used Railtrack lines, and is the standard system on London Underground. If better headways are required then perhaps a 3-aspect system would be necessary.

3-aspect signalling

In order to improve headways the Stop/Repeater signal pairs must be moved closer together as described above. If the pairs move sufficiently, then the Repeater of one signal will be coincident with the Stop Signal in the rear, i.e. signals 3 and 5R will be in the same place.

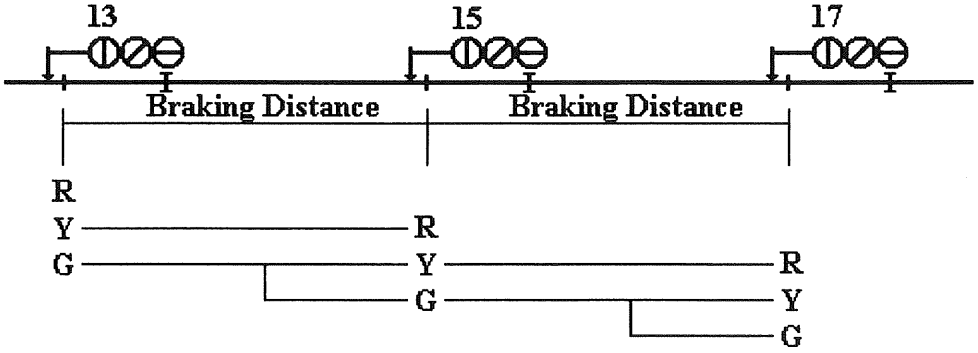
On London Underground this is exactly what happens - two 2-aspect signals on the same post, the upper being the Stop Signal with the Repeater below it.



The repeater works as described in the previous section, except that it is only lit when its adjacent stop signal is showing a green aspect. This stops the signal showing a red stop aspect

and a green repeater aspect at the same time, which would cause unnecessary confusion. The complete signal can, however, show two greens, one on each part of the signal, at the same time if there are no trains in the vicinity.

In these circumstances Railtrack economise and eliminate one of the greens resulting in a 3-aspect signal, showing red, yellow or green. The yellow on one signal warning of a red at the next must be at least the Braking Distance away. Thus consecutive 3-aspect signals must be a minimum of the Braking Distance apart.

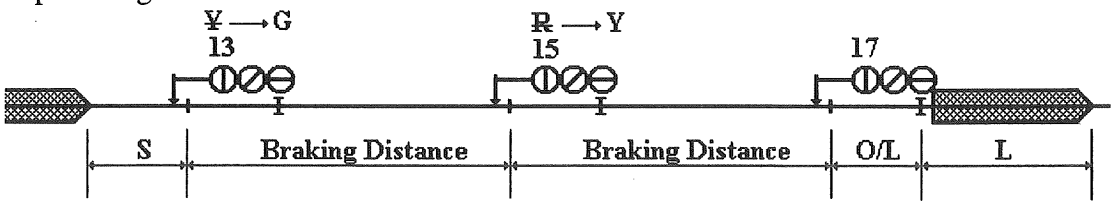


The lower part of this diagram, the *Aspect Sequence Chart*, shows how the aspect of one signal is affected by the signals in advance. The clearing of each signal depends as normal on the condition of the track between it and the next signal and its overlap. The aspect it shows when cleared depends on the aspect of the next signal in advance. Thus 15 will show yellow if 17 is red, or will show green if 17 is showing yellow or green.

Under normal operating conditions any train approaching a red signal will always have received a yellow aspect on the previous signal, which will always be at least braking distance from that red.

One way of measuring the headway is to calculate the minimum possible distance between two trains, with the second train following at such a distance that its driver never sees a restrictive aspect, i.e. only greens.

Aspects of signals will change to a less restrictive aspect when the first train just clears the overlap of a signal.



In the diagram above, the first (right hand) train has just cleared the overlap at 17 signal. At this point in time, 15 will change up from red to yellow, and 13 will change up from yellow to green. The second (left hand) train must be in such a position that signal 13 will not have been seen at yellow.

For the best headway, the signals must be as close as possible, i.e. at Braking Distance apart. A rather crude expression for the headway, the distance between (the fronts of) the trains could be:

$$H_3 = L + O/L + 2BD + S$$

where H_3 is the absolute minimum headway distance for 3-aspect signalling

L is the length of the train

O/L is the overlap length

BD is the braking distance

S is the Sighting Distance, that is the distance of the train from the signal where the aspect can first be discerned

Although it is somewhat clumsy, this expression is useful when used in comparisons with other signalling methods.

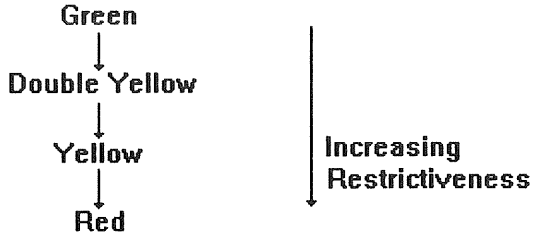
If it is desired to improve the headway even more, then a fourth aspect must be introduced as the signals in a 3-aspect system must not be closer than the Braking Distance apart.

4-aspect signalling

To improve the headway, a fourth aspect is necessary. As all three colours have been used., it is necessary to introduce the concept of more than one lamp being lit at a time. In this situation care must be taken to ensure that a single lamp failure does not result in a dangerous situation..

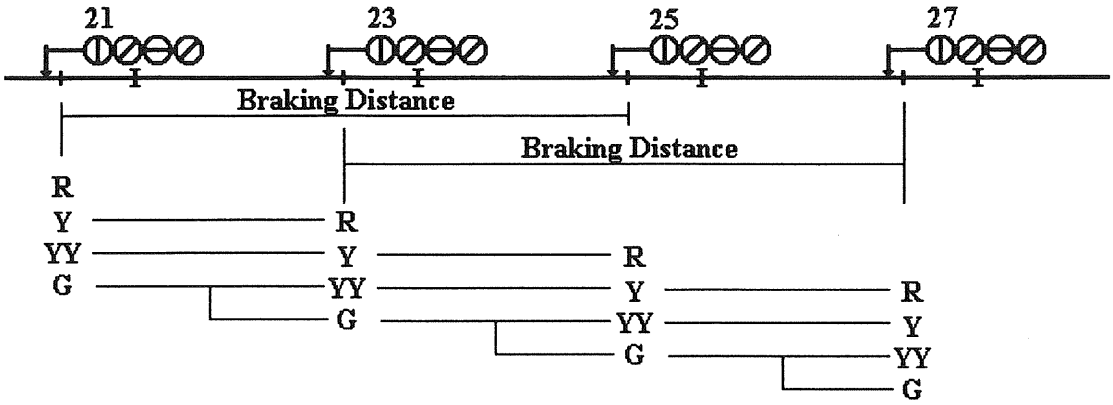
In Belgium there is an aspect with a green and a yellow light together. If the yellow were to fail, then a less restrictive green would remain - a wrongside failure. Complicated circuits are necessary to recognise this failure, and take steps to prevent a wrongside failure. Another example is in France, where there are two different stop aspects, one with a single red, and one with two reds. If one of the reds of the double red aspect was to fail, a less restrictive single red would remain. Special precautions again are necessary to prevent this. These examples show the problems of introducing an aspect with two lamps illuminated at the same time.

The aspect chosen in the UK is two yellows together (*'double yellow'*). This aspect is placed between the green and the yellow aspects in degree of restrictiveness:



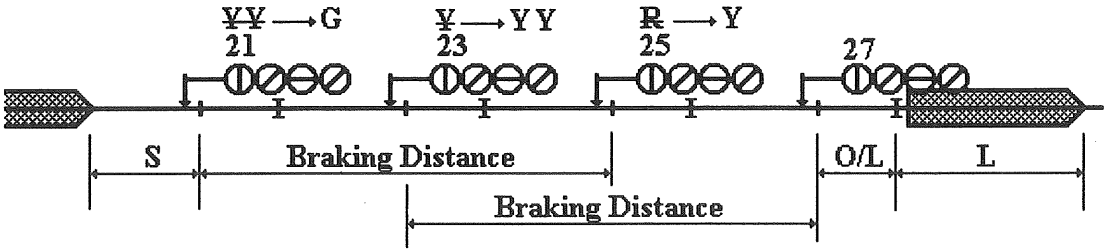
If one of the yellow lamps fails, the resulting single yellow is more restrictive, so no special precautions are necessary.

With 4-aspect signalling a driver first receives a double yellow aspect and then a single yellow before the red is reached. Note that a yellow always reads up to a red (irrespective of whether it is a 2-, 3- or 4-aspect system).



The first restrictive aspect, in this case the double yellow, must as usual be at least at the Braking Distance from the red signal. The closest the signals can be is now half braking distance, i.e. double the number of signals compared with 3-aspect signalling.

A similar exercise to calculate the 4-aspect system headway can be carried out to see the effect of doubling the number of signals.



The best headway for a 4-aspect signalling system, using similar arguments to those used for 3-aspect signalling can be expressed:

$$H_4 = L + O/L + 3/2BD + S$$

where H_4 is the absolute minimum headway distance for 4-aspect signalling.

L, O/L, BD and S are the same as before.

The difference between the two expressions is that the 4-aspect headway distance is reduced by only half the braking distance, that is the difference between 2BD and 3/2BD.

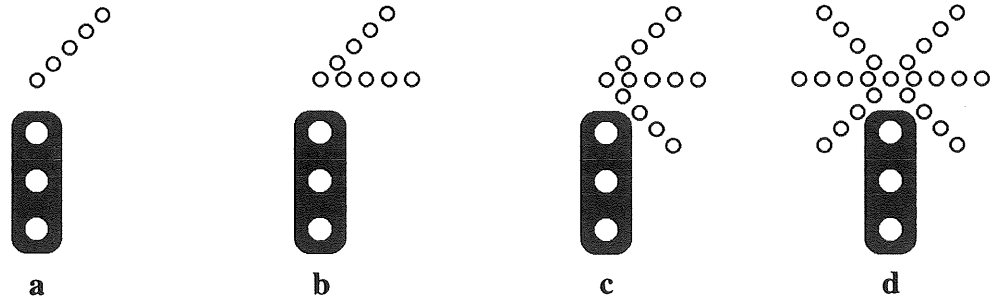
So although the number of signals is doubled, the headway is only improved by one and a half times the Braking Distance, not double. In addition to improving the headway, the 4-aspect system has particular advantages when a mixture of traffic uses the line. High speed heavy trains require a much longer braking distance than slower, lighter suburban trains. The latter can effectively ignore the double yellow aspects and treat them as greens, having sufficient braking to be able to stop from the single yellow aspect. Mixed traffic can thus be kept moving well, without unnecessary delay.

Despite being introduced as long ago as 1926, the 3- and 4-aspect systems have been found very adequate, and have remained virtually unchanged, despite the vast increase in train speeds since that time. The only way to improve headways further would be to improve the braking of trains, taking passenger comfort into consideration.

Junction Signalling

An indication is given to the driver that he is diverging from the 'straight' route by means of a *Junction Direction Indicator (JDI)*, more commonly known as a *Route arm.* This consists of a row of five white lights, mounted above the signal, pointing in the direction of the divergence, to the right or left.

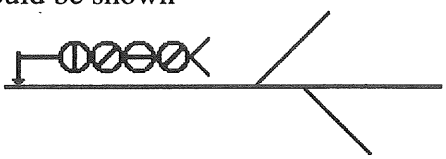
Considering a simple junction with one diverging route to the right, the route arm will point upwards to the right, at 45° (diagram **a** below). If there were two routes to the right then there would be two route arms, with a horizontal row of lights as in diagram **b** below for the right-hand-most route. Note that the innermost lamp is common to the two route arms.



If there were three diverging routes to the right, then a third route arm would be needed, pointing downwards at 45° for the most right-hand route as in diagram **c** above. It is thus possible to have a maximum of six route arms on one signal for the diverging routes, three each side as in diagram **d** above.

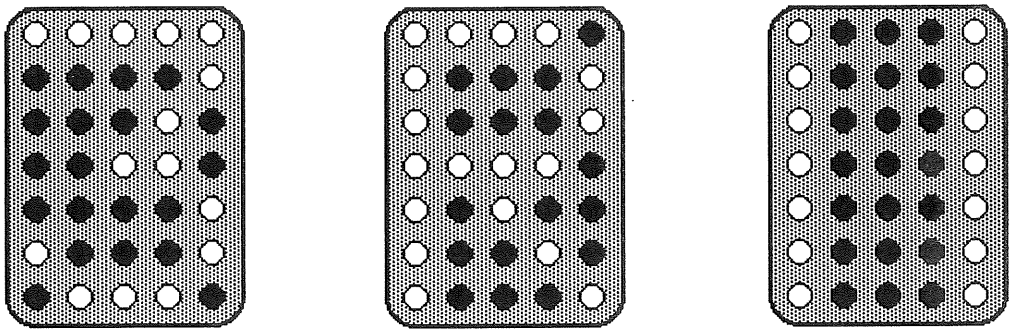
There is no route arm for the straight ahead route in UK practice. The railways in South Africa also use route arms on their signals, but their practice is to additionally have a vertical arm for the 'straight ahead' route.

On track plans the symbol for each route arm fitted is just a straight line, no attempt is made to show the individual lamps involved. A 4-aspect signal with two diverging routes, one to the left, and one to he right would be shown



In the more complex situation at the approach to a terminal station with perhaps twenty platforms, then a different indicator must be used as the route arm idea could not cope. This more versatile indicator is known as a *Theatre Type Indicator* or *Multi Lamp Route Indicator (MLR)*, but if it had been introduced today would probably have been called a dot matrix indicator.

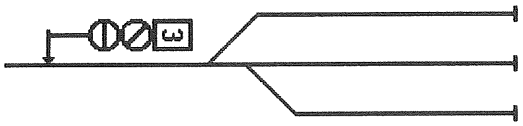
The original design consisted of a matrix of lamps, and by illuminating the correct combination of lamps, the image of a letter or number (or any other shape) could be shown to the driver.



The MLR is only a *short range* signal, that is, it cannot be seen from a great distance, and can only be used where speeds are low. Thus it is ideal for use at the approach to terminal stations. Route arms are *long range* signals and must be visible from the same distance as the coloured light signal to which they are attached.

Current designs of MLRs use fibre optic cables to transfer light from a lamp (one for each indication) to the various 'lenses' at the front of the indicator. This arrangement is much easier to engineer than the original design with a large number of lamps.

The MLR is shown on track plans as a rectangle adjacent to the signal. Inside the rectangle is a number showing the number of different indications that indicator can show, i.e. an indicator mounted above a signal reading into a three platform terminus might be shown.

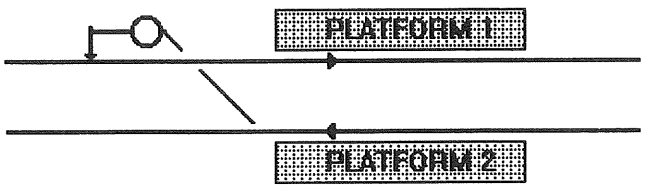


Note that with MLRs an indication would usually be given for every route, and that signals reading into terminal platforms do not have a green aspect.

Fixed Signals

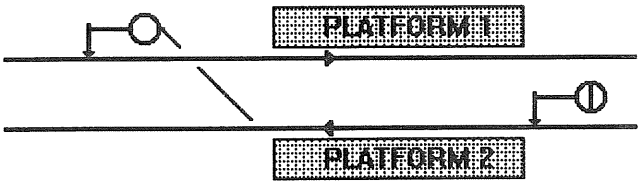
Occasionally signals are needed that only ever show one aspect, i.e. they show the same aspect continuously. These are known as *Fixed* signals.

Consider the situation at a through station where it is required to allow trains entering from the left to reverse in platform 2 to return to where they came.



A signal with a route arm would be provided to signal trains across the crossover into platform 2 (The single circle symbol, with no bar inside represents any signal, the number of aspects not being relevant to the argument). If nothing more was done, there would be

nothing to stop the driver running straight through platform 2 and continuing on the *wrong line* for ever (the normal direction of traffic on the line through platform 2 is from right to left). The solution is to have a *fixed red* signal i.e. permanently showing red, at the end of platform 2. This will stop the train going any further than the end of the platform.



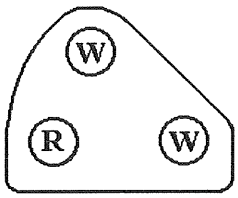
The other example of a fixed signal is the *fixed yellow*. A typical use would be where a little used branch line joins a main line. The actual junction would be protected by a stop signal on each line. On the branch there would be a repeater signal, at least at braking distance from its Stop signal. However, rather than going to the expense of making this a working signal, it might well show yellow permanently, irrespective of the aspect that its parent stop signal is showing. This has the effect of drawing the attention of the driver to the fact that he is 'getting back to civilisation' after having been on the little used branch and possibly not having seen any signals for a considerable period of time.

POSITION LIGHT SIGNALS

Position Light Signals give different aspects by different arrangements or patterns of lights, the colour not necessarily relevant. Some railways use position light signals exclusively - one of the largest American Railways, the Pennsylvania Central, used position light signals for many years. The stop aspect was three white lights arranged horizontally, the proceed aspect by three white lights vertically, and the caution aspect by three white lights in a diagonal arrangement (in fact, very similar to the symbols we use for red, green and yellow lamps).

Shunt Signals

In the UK, position light signals are used for shunt movements, and in fact one of the lamps involved is coloured. The basic signal consists of three lamps roughly in the shape of a triangle.



A shunt signal has only two aspects, stop and proceed. The stop aspect is shown by illuminating the two horizontal lights, the left hand one being red. The proceed aspect is given by illuminating the two diagonal lights, (top left and bottom right) both white. The right hand light is thus continually alight, and is known as the *pivot light*, the other two giving the effect of pivoting around it as they change.

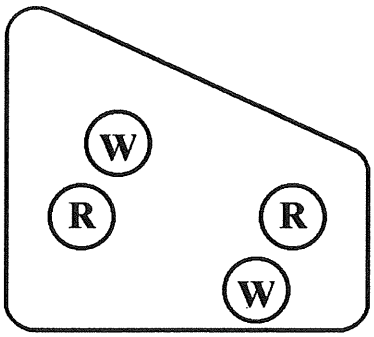
Shunt signals are normally *ground mounted*, i.e. mounted at rail level. Very occasionally, in exceptional circumstances, a pole mounted signal is needed to overcome sighting problems.

The symbol used on track plans does not attempt to show the three lamps in detail, one symbol representing the whole signal, with plain dots for the lamps. As with colour light signals, four varieties of the symbol are required.

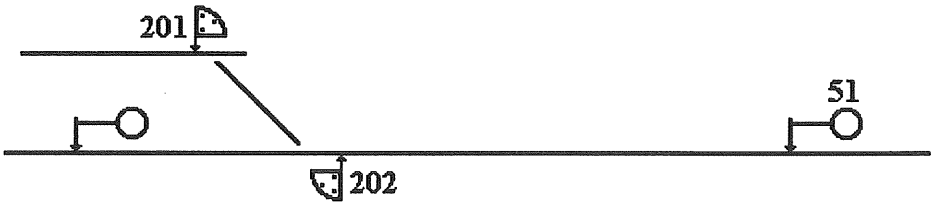


The two left hand symbols are for trains moving from left to right, and the two right hand symbols are for trains moving from right to left.

A new design of shunt signal using fibre optics has the equivalent of four lenses, two white and two red, eliminating the permanently illuminated pivot lamp. However, the indications seen by the driver are slightly different from the traditional signal - the stop aspect is now two reds horizontally, whilst the proceed aspect remains as two whites at an angle.



A typical example of the use of shunt signals is to move trains in and out of sidings. Because of the fears of facing points in the early days of railways, most sidings are connected to the mainline in a trailing direction, which means that trains have to back into the siding from the main line.



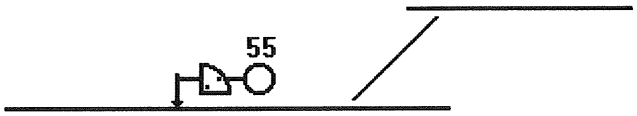
The train to enter the siding would be brought to a halt with a red aspect on 51 signal, having passed, but only having seen the back of, 202 signal. Shunt signal 202 would then be cleared for the train to back into the siding. Hopefully the distance between 202 and 51 signals should be sufficient to accommodate the longest train to use the siding facilities. Signal 201 is provided to signal trains out of the siding back onto the main line to continue on their journey under the control of 51 signal. On a big yard the exit signal might well be a colour light main signal rather than a shunt signal.

Stencil Indicators

Shunt signals frequently have several routes and the destinations need to be indicated to the driver. A simple display, a *stencil indicator*, is normally used. It consists of a plate of glass, with appropriate legends etched on the back. The legend is not visible until illuminated from the rear by a lamp.

Subsidiary Signals

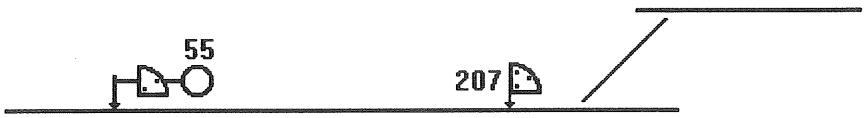
If the connection into a yard is a facing connection, the situation may become somewhat more complicated. There is a rule that main signals may only read up to another main signal (or the buffers at a passenger station). Thus the entry into a simple siding must be controlled by a shunt signal, not a main. The solution is to mount the shunt signal on the same post as the stop signal. In this situation, the signal is known as a *subsidiary signal* and the red light is dispensed with. The subsidiary signal is thus normally dark and is only lit (with the two white lights) when it is cleared.



To enter this siding the train is brought to a halt with a red aspect on 55 main signal. When the train is proved to be stationary the subsidiary signal is cleared, instructing the driver to pass the red aspect and enter the siding. Note that the driver is not being asked to pass the red at speed, only from a stationary position. Any train passing along the main line does not see any red light on the subsidiary signal.

Preceding Shunt Signals

In some circumstances it may not be possible to position the signal close to the entry, and an additional signal, 207, might be needed, especially if the main line has other connections not shown in the diagram.

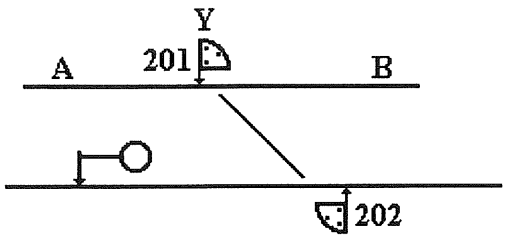


However, any train moving along the main line would see the aspect of 207, including the red light. As explained earlier, it is not acceptable to ask drivers to pass red lights at speed, and special arrangements have to be made. These arrangements ensure that if the signaller wishes to route a train along the main line from 55 signal, then the shunt signal 207 will also be cleared, thus eliminating the sighting of the red light. The interlocking system does this automatically, the signaller need take no extra actions. Such a signal is known as a *Preceding Shunt* Signal.

Signal 55 still needs a subsidiary signal, as a main signal is not allowed to read up to a shunt signal. Main signals are allowed, however, to 'read over' a preceding shunt signal to another main signal.

Yellow Shunts

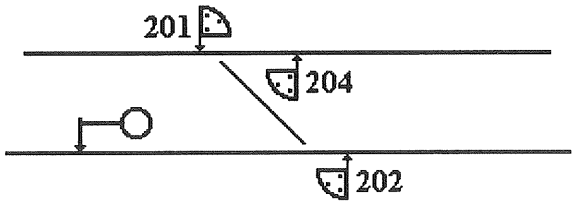
Consider the situation at slightly more complicated yard.



Trains might wish to shunt backwards and forwards in the yard between A and B. However, the signaller should not need to be bothered for each of these movements - his job should be concentrated on the main line. The signal 201 may be fitted with a yellow lamp instead of the red, and this allows the driver to pass the signal when showing the yellow/white aspect as long as he is not being routed on to the main line. Thus the signaller need only worry about signal 201 when a train needs to leave the yard onto the main line.

Opposing Locking Omitted Signals

If the yard was busier still, with more than one train in at a time, then a second signal, protecting the points would be needed.



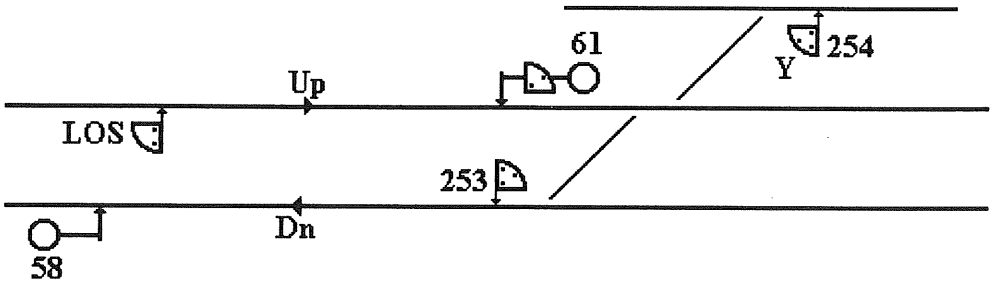
Again the signaller does not want to be worried by movements within the yard. In this situation it would be expected that a shunter would be supervising movements. Under normal circumstances such opposing signals would be locked against each other so they could not both show a proceed aspect at the same time. Special arrangements can be made in these circumstances so that both signals in the yard <u>can</u> show a proceed aspect at the same time. This is known as *Opposing Locking Omitted* and can only be applied whilst the points are <u>not</u> set towards the main line.

The Limit of Shunt signal

The last type of shunt signal is the *Limit of Shunt (LOS)* signal, the equivalent in shunt terms of a fixed red main signal. It has only two lamps, both red, in the two horizontal positions of the signal. Like a fixed red, the LOS is illuminated permanently. The symbol just has the two dots in the relevant positions.

Shunt Summary

The diagram below shows a typical use of four of the different types of shunt signals described.

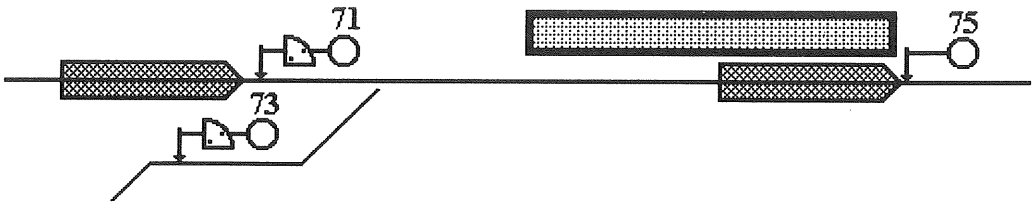


Entry to the siding from the Up line would be by means of the subsidiary signal on 61 signal, whilst entry from the down line is taken care of by the shunt signal 253. The exit signal from the yard, 254, is a yellow shunt which reads straight out to signal 58 on the Down line, or to the LOS on the Up line. In the latter case the train would continue in the Up direction, 61 main signal having been cleared.

Call-On moves

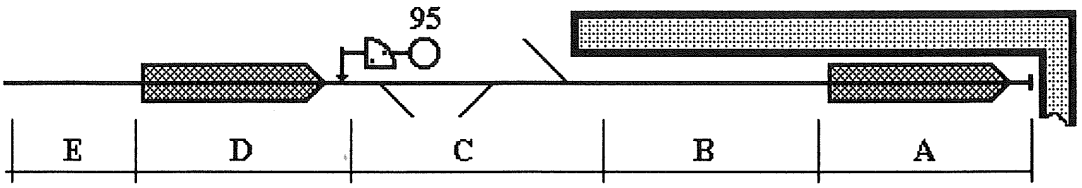
All the shunt signal moves described above are restricted to non passenger carrying trains, i.e. light engines, empty carriages and freight trains. However there is one particular case, described below, where passenger carrying trains are controlled by shunt signals.

There are many places where there is a requirement to bring two trains from different starting points into the same platform to join them together to then form one train to continue on its journey. This sort of move can only be carried out with multiple unit stock without separate locomotives.



The diagram shows a station with the first train sitting in the platform at 75 signal. This train probably approached the station from 73 signal. A second train, approaching 71 signal is required to enter the occupied platform, to be joined on the back of the first train. The first train must be holding signal 71 at red, so that the second train cannot reach the platform. A subsidiary signal must be fitted to 71 to allow a *call-on move* into the occupied platform. Even though the signaller has set the route from 71 to 75 the aspect of 71 remains at red (Because of the first train anyway). However, when the second train is proved to have come to a halt at 71, the subsidiary signal clears (with no further action by the signaller) to instruct the driver to pass the red and cautiously enter the occupied platform.

Call-ons are also found in terminal stations to allow two (or more in some cases) trains into the same platform, not necessarily to join them together. Suppose the track is divided into five section as shown, A, B, C, D and E.



When the signaller requests the route for the first train, the interlocking will check that Sections A, B and C are not occupied by any vehicles and allow 95 signal to clear to a yellow aspect. When the signaller requests the route for the second train the interlocking will detect that Section A is occupied by the first train and use the Call-on subsidiary signal for the second, when it is proved to be stationery on Section D. This second train then moves cautiously into the platform, drawing up close to the first train.

However, there would be problems if the second train was too long for the available space left in the platform. The back of the train would hang out over the points in the station throat blocking a large section of the station. A technique, known as *Lime Street Control*, is available to prevent this problem happening. (Lime Street is a station in Liverpool where, presumably, the technique was first tried.) The crucial part of the checking process is to make Section D the same length as Section B. When the interlocking tries for the Call-on route, it will only work if Section E is proved to be clear. With the train standing at 95 signal occupying Section D, then if Section E is also occupied, the train must be longer than Section D, which means it must also be longer than Section B (as Sections B and D are deliberately the same length). The train is thus proved to be too long to fit in the platform, the Call-on will not clear, and the signaller will have to find somewhere else for the train to go.

The conditions for the main signal to clear are therefore Sections A, B and C all clear. For the Call-on to clear, Sections B, C and E must be clear with A and D occupied.

SPAD SIGNALS

A special signal has been introduced to indicate to drivers that they have just passed a signal showing a red aspect. The number of Signals Passed At Danger (*SPAD*) has risen to an unacceptable level, so a SPAD signal is installed in situations where such passing of a red signal would result in a high risk of an incident. It is usually installed a few metres ahead of the signal with which it is associated. The SPAD signal consists of three red lamps arranged vertically, but only shows one aspect, it being normally dark. When a SPAD is detected, by the overlap track at a signal being occupied whilst the signal is showing a red aspect, the SPAD signal is illuminated, showing a red aspect on the middle lamp, with a flashing red above and below it.